In [1]:

```
!pip install apyori -q

import pandas as pd
import matplotlib.pyplot as plt
from apyori import apriori

df = pd.read_csv("Market_Basket_Optimisation.csv",
header=None)

print(df.head())
print("Shape:", df.shape)

transactions = []

for i in range(len(df)):
    transactions.append(
        [str(df.values[i, j]) for j in range(df.shape[1])
         if str(df.values[i, j]) != "nan"]
    )

rules = apriori(
    transactions,
    min_support=0.003,
    min_confidence=0.2,
    min_lift=3,
    min_length=2
)

rules = list(rules)

print("Number of rules:", len(rules))

result = []

for rule in rules:
    items = list(rule.items)
    for ordered_stat in rule.ordered_statistics:
        result.append([
            ", ".join(ordered_stat.items_base),
            ", ".join(ordered_stat.items_add),
            rule.support,
            ordered_stat.confidence,
            ordered_stat.lift
        ])

rules_df = pd.DataFrame(
    result,
    columns=["Antecedent", "Consequent", "Support",
"Confidence", "Lift"]
)
```

```
rules_df = rules_df.sort_values(
    by="Confidence", ascending=False
).reset_index(drop=True)

print("\nAssociation Rules:")
print(rules_df.head(10))

plt.figure(figsize=(10,6))
plt.scatter(
    rules_df["Support"],
    rules_df["Confidence"],
    c=rules_df["Lift"],
    cmap="viridis",
    s=50
)
plt.colorbar(label="Lift")
plt.xlabel("Support")
plt.ylabel("Confidence")
plt.title("Association Rules")
plt.show()

# Increasing minimum confidence
for confidence in [0.3, 0.4, 0.5, 0.6, 0.7]:
    r = list(apriori(
        transactions,
        min_support=0.003,
        min_confidence=confidence,
        min_lift=3,
        min_length=2
    ))
    print("Minimum Confidence:", confidence,
          "  Number of Rules:", len(r))
```

```
  Preparing metadata (setup.py) ... [?25l[?25hdone
  Building wheel for apyori (setup.py) ... [?25l[?
25hdone
             0          1           2                3
4   \
0         shrimp    almonds     avocado    vegetables
mix  green grapes
1        burgers  meatballs        eggs
NaN           NaN
2        chutney        NaN         NaN
NaN           NaN
3         turkey    avocado         NaN
NaN           NaN
4  mineral water       milk  energy bar  whole wheat
rice     green tea

                5     6              7             8
9   \
0  whole weat flour  yams  cottage cheese  energy drink
tomato juice
1              NaN   NaN             NaN           NaN
```

```
NaN
2               NaN   NaN             NaN           NaN
NaN
3               NaN   NaN             NaN           NaN
NaN
4               NaN   NaN             NaN           NaN
NaN

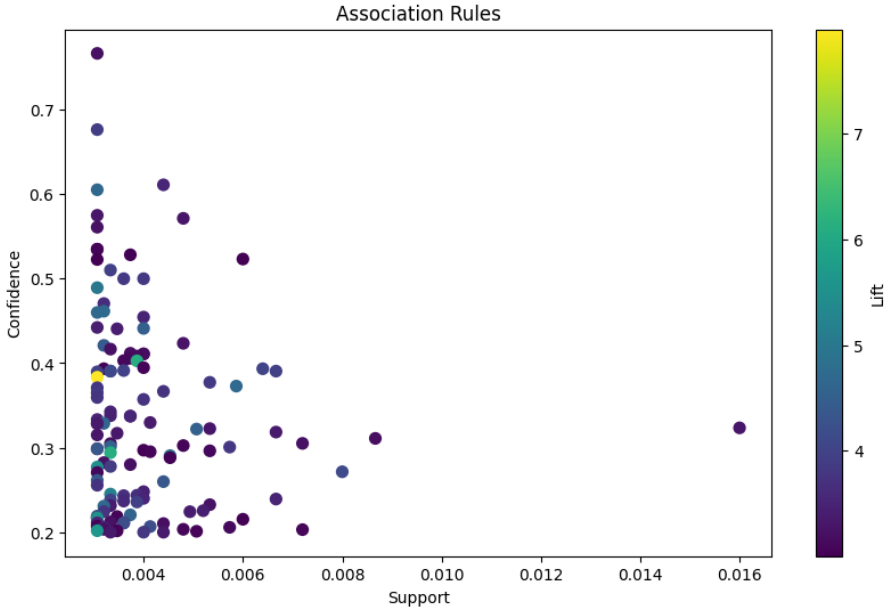
               10         11     12     13
14      15  \
0  low fat yogurt  green tea  honey  salad  mineral
water  salmon
1             NaN        NaN    NaN    NaN
NaN     NaN
2             NaN        NaN    NaN    NaN
NaN     NaN
3             NaN        NaN    NaN    NaN
NaN     NaN
4             NaN        NaN    NaN    NaN
NaN     NaN

                  16               17       18
19
0  antioxydant juice  frozen smoothie  spinach  olive
oil
1                NaN              NaN      NaN
NaN
2                NaN              NaN      NaN
NaN
3                NaN              NaN      NaN
NaN
4                NaN              NaN      NaN
NaN
Shape: (7501, 20)
Number of rules: 80

Association Rules:
                                  Antecedent
Consequent   Support  \
0              frozen vegetables, soup, milk  mineral
water  0.003066
1                       cereals, ground beef
spaghetti  0.003066
2                        tomatoes, olive oil
spaghetti  0.004399
3     mineral water, frozen vegetables, soup
milk  0.003066
4                  tomato sauce, ground beef
spaghetti  0.003066
5                   cooking oil, ground beef
spaghetti  0.004799
6       mineral water, ground beef, tomatoes
spaghetti  0.003066
7       frozen vegetables, milk, ground beef
```

```
spaghetti  0.003066
8  chocolate, frozen vegetables, ground beef
spaghetti  0.003066
9                              red wine, eggs
spaghetti  0.003733
```

	Confidence	Lift
0	0.766667	3.216312
1	0.676471	3.885303
2	0.611111	3.509912
3	0.605263	4.670863
4	0.575000	3.302508
5	0.571429	3.281995
6	0.560976	3.221959
7	0.534884	3.072100
8	0.534884	3.072100
9	0.528302	3.034297




```
Minimum Confidence: 0.3   Number of Rules: 51
Minimum Confidence: 0.4   Number of Rules: 30
Minimum Confidence: 0.5   Number of Rules: 14
Minimum Confidence: 0.6   Number of Rules: 3
Minimum Confidence: 0.7   Number of Rules: 1
```

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